

AI Travel Product Management System

Technical Assessment (24 Hours)

Overview

Develop a simple AI-powered Travel Product Management System that enables users to create, manage, and search travel products using the OpenAI API. The objective is to assess your full-stack development skills, CRUD implementation, API integration, AI prompt engineering, and application design.

Duration

24 Hours

Objective

Build a web application that allows users to:

- Create travel products
 - Manage products through CRUD operations
 - Generate product information using AI
 - Search products using natural language
 - Automatically hide expired products based on inventory validity
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Technology Stack

You may use any technology stack of your choice.

Examples:

- Frontend: React, Angular, Vue
 - Backend: ASP.NET Core, Node.js, Laravel, Django
 - Database: SQL Server, MySQL, PostgreSQL
 - AI: OpenAI API
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Functional Requirements

1. Product Management (CRUD)

The application should allow users to:

- Create a Product
- View Products
- Update Products
- Delete Products

Product Fields

Field	Required
Product Name	Yes
Destination	Yes
Category	Yes
Description	Yes
Price	Yes
Inventory Count	Yes
Valid From	Yes
Valid Until	Yes
Status (Active/Inactive)	Yes

2. AI Product Generation

Provide an input where users can describe a product in natural language.

Example Prompt

Create a Dinner Buffet at Cinnamon Grand Colombo available until the end of this month.

Using the OpenAI API, automatically generate:

- Product Name
- Product Description
- Highlights
- Inclusions
- Suggested Category
- Tags

The generated details should populate the product form automatically, allowing the user to review and save the product.

3. AI Search

Allow users to search products using natural language.

Example Searches

- Show me dinner buffets in Colombo
- Show active family packages
- Show products below LKR 10,000
- Show airport transfer services

The application should use OpenAI to interpret the user's request and apply the appropriate filters before returning matching products.

4. Product Validity

Products should only be visible if they are within their validity period.

Business Rule

If the **Valid Until** date has passed:

- The product should not appear in search results.
- The product should automatically be considered expired.

Example

Product

Dinner Buffet – Cinnamon Grand Colombo

Valid Until: 31 July 2026

Current Date: 1 August 2026

Expected Result

The product should not appear in the product listing or AI search results.

5. Dashboard

Display the following summary:

- Total Products
 - Active Products
 - Expired Products
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Database

A simple database structure is sufficient.

Suggested Product Table

- ProductId
 - ProductName
 - Destination
 - Category
 - Description
 - Price
 - InventoryCount
 - ValidFrom
 - ValidUntil
 - Status
 - CreatedDate
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Evaluation Criteria

Criteria	Marks
CRUD Functionality	20
Database Design	15
OpenAI Integration	25
AI Search	20
UI / User Experience	10
Code Quality	10
Total	100

Bonus Features (Optional)

- Product image generation using AI
 - Responsive UI
 - Export products to PDF or Excel
 - Docker deployment
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Submission Requirements

Please submit:

- Source Code
 - Database Script
 - README with setup instructions
 - `.env.example` file for environment configuration (without API keys)
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What We Will Evaluate

- Correct implementation of CRUD operations
- Effective integration with the OpenAI API
- Logical handling of product validity and inventory
- Clean code structure and application architecture
- User-friendly interface
- Overall quality and completeness of the solution